

Part I – Contributions to research and development

a. Articles published or accepted in peer-reviewed journals

Tomlinson, K. W., Yu, F., **Wang, X.**, Yao, X., Yu, C. C., Charles-Dominique, T., ... & G  lin, U. (2025). The macroecology of spines on woody plants. *Biological Reviews*, 100(6), 2396-2419.

Wang, X., Fan, H., Phoncharoen, W., G  lin, U., & Tomlinson, K. W. (2023). Leaf chemistry of architecturally defended plants responds more strongly to soil phosphorus variation than non-architecturally defended ones. *Physiologia Plantarum*, 175(1), e13856.

Wang, X., & Ib    ez, I. (2022). The contrasting effects of local environmental conditions on tree growth between populations at different latitudes. *Forests*, 13(3), 429 (Master's thesis)

b. Other peer-reviewed contributions

c. Non-peer-reviewed contributions

Wang, X., Van der Meersch, V., & Wolkovich, E. M. (2026) Fine Balance: Exploring the Interplay of Reproductive Strategies and Growth Dynamics in Trees, poster presented at the StanCon 2026: International Conference on Bayesian Inference and Probabilistic Programming held on August 19 at Uppsala University, Uppsala, Sweden

Wang, X. (2022) Leaf chemistry of architecturally defended plants responds more strongly to soil phosphorus variation than non-architecturally defended ones, oral presentation at the 2022 Xishuangbanna Tropical Botanical Garden Annual Academic Conference held in Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences, Xishuangbanna, China

Wang, X., Wang, H., & Liao, A. (2018) Environmental Characteristics of Seagrass Beds *Halophila beccarii* at Tangjiawan, Zhuhai, poster presentation at the Sixth Undergraduate Science and Technology Poster Presentation Conference at BNU-HKBU United International College, Zhuhai, China

Wang X., Roybal, O., Hospeti, L., & Lenihan, M. (2017) The influence of sterilization on germination of six *Asclepias* species, poster presentation at the 11th Annual Summer Undergraduate Research Conference at Trinity University, San Antonio, United States

d. Technology transfer

e. Contributions resulting from your participation in industrially relevant R&D activities

f. Awarded and submitted patents and copyrights

Part II – Most significant contributions to research and development

Investigate climate impact on growth My master's project focused on how environmental changes influence the growth of the same species across different latitudes. Our results showed that even populations located near the northern edge of their distribution are already experiencing climate-related stress and may struggle to persist. We used models to predict future performance under climate change, and the sugar maples are projected to decline. This work, published in *Forests*, emphasized the importance of higher spatial resolution in predicting species responses to climate change.

Collaborative work in macroecology As a research assistant to Dr. Kyle Tomlinson, I contributed to a global study on the distribution and evolution of spiny plants. I led data cleaning and spine identification and engaged with international collaborators, including contacting experts to verify spine morphologies. This work, published in *Biological Reviews*, provided a global synthesis of the distribution and ecology of spiny plants.

Promote endangered seagrass conservation My undergraduate senior project focused on studying the only known patch of endangered *Halophila beccarii* seagrass in Zhuhai, which at the time was not under any protection. Towards the end of my project, the local government proposed a beach restoration plan that involved relocating sand to the original rocky shoreline, which directly threatens the seagrass habitat. We sent the central government a letter highlighting the ecological importance of the area, supported by field data showing higher biodiversity in the seagrass bed than in nearby bare ground. We subsequently collaborated with research teams organized by the government, and the seagrass patch has now been officially designated as a protected area.

Part III – Applicant’s statement

Research Experience

I began my research training in ecology through my undergraduate final project, where I investigated the community structure of the endangered seagrass *Halophila beccarii* and its interactions with algae under different environmental and nutrient conditions. This project provided my first experience with ecological fieldwork, study design, data collection, and statistical analysis, and established my interest in plant–environment interactions.

During my master’s degree, I expanded this interest to terrestrial ecosystems and investigated how climate influences forest dynamics. My thesis examined how temperature and precipitation affect annual growth of two common maple species across latitudinal gradients in Michigan using dendrochronology and Bayesian modeling. This work developed my expertise in tree-ring analysis, statistical modeling in R, and ecological interpretation of long-term environmental data, while providing experience independently developing and addressing a quantitative research question.

Following my master’s degree, I worked as a research assistant in a community ecology group, where I supported field research, plant specimen identification, data management, and statistical analysis. I also assisted students with fieldwork and research projects, strengthening my ability to work collaboratively, communicate scientific methods, and contribute to multiple research projects.

Together, these experiences gave me a broad foundation in field ecology, quantitative analysis, and forest ecology and prepared me to undertake independent doctoral research on the ecological and evolutionary mechanisms underlying tree reproduction and masting.

Relevant activities

During my undergraduate degree, I was selected as the only junior student in my cohort to participate in a fully funded summer research program in Texas. During my master’s degree, I further broadened my ecological training through additional coursework and developed quantitative expertise through workshops in advanced statistics and Bayesian modeling. I applied these skills to ecological datasets throughout my graduate research.

I have also actively contributed to research training and mentorship. As a research assistant, I organized free R workshops for undergraduate students to develop their coding and data-analysis skills. During my PhD, I have trained and mentored several WorkLearn students, providing guidance on field methods, data analysis, and research projects. These experiences have strengthened my ability to communicate technical concepts, support students with different levels of research experience, and contribute to an inclusive research environment.

Beyond research and teaching, I have participated in science outreach and academic service. I worked with the Zhuhai Bird Watching Association to organize public education activities about seagrass habitats. During my PhD, I also volunteered for the EcoEvo Annual Retreat, helping coordinate speakers and organize scientific talks. These activities have allowed me to communicate ecological research to broader audiences and contribute to my academic community.